

1. Import

```
import matplotlib.pyplot as plt
```

Standard convention: plt

Optionally for interactive Jupyter plots:

```
%matplotlib inline
```

2. Basic Plotting

Function Description Example

plt.plot() Line plot plt.plot(x, y)

plt.scatter() Scatter plot plt.scatter(x, y)

plt.bar() Vertical bar chart plt.bar(categories, values)

plt.barh() Horizontal bar chart plt.barh(categories, values)

plt.hist() Histogram plt.hist(data, bins=10)

plt.boxplot() Box plot plt.boxplot(data)

plt.pie() Pie chart plt.pie(values, labels=categories)

3. Plot Customization

Function / Parameter Description Example

plt.title() Add title plt.title("My Plot")

plt.xlabel() X-axis label plt.xlabel("X-axis")

plt.ylabel() Y-axis label plt.ylabel("Y-axis")

plt.legend() Show legend plt.plot(x, y, label="Line1"); plt.legend()

plt.grid() Show grid plt.grid(True)

color Line/bar color plt.plot(x, y, color='red')

linestyle Line style plt.plot(x, y, linestyle='--')

marker Marker style plt.plot(x, y, marker='o')

alpha Transparency plt.scatter(x, y, alpha=0.5)

figsize Figure size plt.figure(figsize=(10,5))

4. Multiple Plots

```
plt.plot(x1, y1, label="Line1")
```

```
plt.plot(x2, y2, label="Line2")
```

```
plt.legend()
```

```
plt.show()
```

Subplots:

```
fig, ax = plt.subplots(1, 2, figsize=(12,4)) # 1 row, 2 cols
```

```
ax[0].plot(x, y)
```

```
ax[1].bar(categories, values)
```

5. Histograms & Density Plots

```
plt.hist(data, bins=20, color='blue', alpha=0.7)
```

```
plt.show()
```

Density plot approximation: plt.hist(data, bins=20, density=True)

6. Scatter Plots

```
plt.scatter(x, y, color='red', s=50, alpha=0.5) # s=size
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
plt.title("Scatter Plot")
```

7. Bar Charts

```
plt.bar(categories, values, color='green', alpha=0.6)
```

```
plt.barh(categories, values, color='orange')
```

Stacked bar chart:

```
plt.bar(categories, values1)
plt.bar(categories, values2, bottom=values1)
8. Pie Charts
plt.pie(values, labels=categories, autopct='%1.1f%%', startangle=90)
plt.axis('equal') # Make circle
9. Box Plots
plt.boxplot(data)
plt.title("Box Plot")
10. Save Plot
plt.savefig("plot.png", dpi=300, bbox_inches='tight')
11. Advanced Customizations
```

Log scale: `plt.yscale('log')`

Rotate xticks: `plt.xticks(rotation=45)`

Figure size & DPI: `plt.figure(figsize=(8,6), dpi=150)`

Multiple lines with styles:

```
plt.plot(x, y1, 'r--', label='Line1')
plt.plot(x, y2, 'b-o', label='Line2')
```

12. Tips & Rules

Large datasets ☒ Use $\alpha < 1$ for scatter, smaller markers, avoid too many lines.

Always label axes & add title.

Use `fig, ax = plt.subplots()` for better control.

For statistical plots consider seaborn on top of matplotlib for easier visualization.